

Jyotirmaya Ijaradar, M.Sc.

Research Associate

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Professional Summary

Research Associate at TU Dresden with expertise in machine learning, deep learning, and data-driven intelligent transportation systems. Experienced in developing scalable AI solutions for traffic prediction, anomaly detection, and large-scale sensor data analysis. My work includes building end-to-end frameworks for real-world applications, combining research and software development. I am passionate about applying advanced technologies to solve urban challenges, continuously learning new methods, and contributing to innovative and socially impactful solutions.

Research Interests

Language Models for Traffic Applications

Data-driven Traffic Optimisation

AI for Autonomous Mobility

AI for Traffic Big Data

Professional Experience

Jul 2023 – Present **Research Associate**

Chair of Traffic Process Automation, TU Dresden Dresden, Germany
Conducting research in traffic data analytics, machine learning, and intelligent transportation systems

Feb 2023 – May 2023 **Student Research Assistant (SHK)**

Fraunhofer Institute for Integrated Circuits (IIS) Dresden, Germany
Conducted research on industrial wireless network monitoring

Jun 2020 – March 2023 **Student Research Assistant (SHK)**

Leibniz Institute of Polymer Research Dresden (IPF) Dresden, Germany
Conducted research on soft robotics and triboelectric nanogenerators

Jan 2021 – Jan 2022 **Student Research Assistant (SHK)**

National Center for Tumor Diseases (NCT) Dresden Dresden, Germany
Developed robot motion planning solutions using ROS and MoveIt for surgical robotics applications

Research Projects

National Park Parking Demand Prediction

2025 – Present

Developing ML models for parking demand forecasting at Sächsische Schweiz National Park

Traffic Insight 360

2024 – Present

Building comprehensive traffic data analytics tools for research, education and products

Teaching Experience

2024 – Present **Course Instructor / Teaching Assistant**
TU Dresden, Faculty of Transport and Traffic Sciences Dresden, Germany
Applied Robotics and Control for Cooperative and Autonomous Mobility: Teaching robotics fundamentals, computer vision, and applications in autonomous vehicles and cooperative mobility systems

Education

2023 **Master of Science in Computational Modeling and Simulation**
Technische Universität Dresden Dresden, Germany
Thesis: Imputation of Missing Packets in Passively Monitored Wireless Network Traffic – A Deep Learning Approach (conducted at Fraunhofer IIS)
Key Courses: Machine Learning I & II, Computer Vision, Data Visualization, Computer Graphics, Statistics and Experimental Design, Computer and Robot-assisted Surgery

2018 **Bachelor of Technology in Electronics and Communication Engineering**
Centurion University of Technology and Management Odisha, India
Key Courses: Data Structures, Embedded Systems, Digital Signal Processing, VLSI Design and Simulation, Data Communication and Networking, Data Mining, Database Systems

Publications

Journal Articles

- 1 A. Richter, **J. Ijaradar**, U. Wetzker, V. Jain, A. Frotzscher, “A Survey on Multivariate Time Series Imputation using Adversarial Learning,” *IEEE Access*, vol. 12, pp. 148167–148189, 2024. (8 citations)
- 2 P. Yan, Z. Li, **J. Ijaradar**, S. Pape, M. Körner, M. Wang, “An Imputation-Enhanced Hybrid Deep Learning Approach for Traffic Volume Prediction in Urban Networks: A Case Study in Dresden,” *Data Science for Transportation*, vol. 6(3), p. 22, 2024. (9 citations)
- 3 S. Hait, L. Kumar, **J. Ijaradar**, A.K. Ghosh, D. De, J. Chanda, P. Ghosh, et al., “Unlocking the Potential of Lignin: Towards a Sustainable Solution for Tire Rubber Compound Reinforcement,” *Journal of Cleaner Production*, vol. 470, 143274, 2024. (18 citations)

Conference Papers

- 4 **J. Ijaradar**, S. Pape, C. Tan, M. Körner, M. Wang, “Data-Driven Anomaly Detection in Urban Traffic Data: A Deep Learning Approach,” *9th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)*, 2025.
- 5 S.K. Sikder, **J. Ijaradar**, H. Li, H. Omrani, “Transition of Car-based Human-mobility in the Pandemic Era: Data Insight from a Cross-border Region in Europe,” *arXiv preprint arXiv:2509.05166*, 2025.

- 6 A. Richter, **J. Ijaradar**, U. Wetzker, “Iterative Imputation of Incomplete Wireless Network Traffic using Adversarial Learning and a Two-Stage Approach,” *3rd International Conference on Artificial Intelligence for Internet of Things (AIIoT)*, 2024. (1 citations)
- 7 S. Teichmann, L. Ninnemann, J. Schanze, **J. Ijaradar**, H. Herold, “Local Differences in Feature Importance for the Prediction of PM10, O3 and NOx using Dresden as Example,” *28th AGILE Conference on Geographic Information Science – Poster Track*, Dresden, Germany, 2025.

Posters & Presentations

- 1 N. Xie, **J. Ijaradar**, L. Wei, S. Pape, M. Körner, M. Wang, “Short-term Impacts of Carola Bridge Collapse on Road Traffic,” *13th Symposium of the European Association for Research in Transportation (hEART)*, Munich, Germany, 2025.
- 2 **J. Ijaradar**, M. Körner, S. Pape, M. Wang, “LSUTDI: A Hybrid Deep Learning Model for Imputing Large-Scale Urban Traffic Data,” *IOER Conference 2024*, Dresden, Germany, 2024.
- 3 **J. Ijaradar**, S. Wunsch, M. Wang, “Estimating Dwell Time for Public Transport,” *COSEAL Workshop 2024*, Dresden, Germany, 2024.

Technical Skills

Programming | Python C C++ R MATLAB JavaScript Vue.js
Deep Learning | PyTorch TensorFlow Keras
Data Science | Pandas NumPy Scikit-learn SciPy
Databases | PostgreSQL MySQL DuckDB
Visualization | Tableau Pentaho Matplotlib Seaborn
Robotics | ROS

Certifications

Coursera Introduction to Self-Driving Cars – University of Toronto
 State Estimation & Localization for Self-Driving Cars – University of Toronto
Udemy Complete Self-Driving Car Course: Applied Deep Learning
 Machine Learning A-Z: Python and R in Data Science
 PyTorch for Deep Learning with Python Bootcamp
 Complete TensorFlow 2 and Keras Deep Learning Bootcamp

Languages

Bengali	Native	English	Professional (B2)
Hindi	Fluent (C1)	Odia	Conversational (B2)
German	Elementary (A2)		